Q1 - 2024 (04 Apr Shift 2)

A parallel plate capacitor of capacitance 12.5pF is charged by a battery connected between its plates to potential difference of 12.0 V. The battery is now disconnected and a dielectric slab $(\epsilon_r = 6)$ is inserted between the plates. The change in its potential energy after inserting the dielectric slab is _____ 10^{-12} J.

Q2 - 2024 (05 Apr Shift 1)

Three capacitors of capacitances 25μF, 30μF and 45μF are connected in parallel to a supply of 100 V. Energy stored in the above combination is E. When these capacitors are connected in series to the same supply, the stored energy is $\frac{9}{x}$E. The value of x is _____.

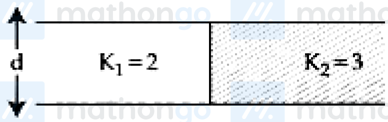
Q3 - 2024 (05 Apr Shift 1)

The electric field between the two parallel plates of a capacitor of 1.5μF capacitance drops to one third of its initial value in 6.6μs when the plates are connected by a thin wire.

The resistance of this wire is _____Ω. (Given, $\log 3 = 1.1$)

Q4 - 2024 (06 Apr Shift 2)

A capacitor of 10μF capacitance whose plates are separated by 10 mm through air and each plate has area 4 cm^2 is now filled equally with two dielectric media of $K_1 = 2$, $K_2 = 3$ respectively as shown in figure. If new force between the plates is 8 N. The supply voltage is _________ $\times 10^{-4}$ V.



(we modified language of question to make it correct)

Q5 - 2024 (08 Apr Shift 2)

A capacitor has air as dielectric medium and two conducting plates of area 12 cm^2 and they are 0.6 cm apart. When a slab of dielectric having area 12 cm^2 and 0.6 cm thickness is inserted between the plates, one of the conducting plates has to be moved by 0.2 cm to keep the capacitance same as in previous case. The dielectric constant of the slab is : (Given $\epsilon_0 = 8.834 \times 10^{-12}\ \mathrm{F/m}$)

(1) 1

(2) 1.33

(3) 0.66

(4) 1.50

Q6 - 2024 (09 Apr Shift 1)

A bulb and a capacitor are connected in series across an ac supply. A dielectric is then placed between the plates of the capacitor. The glow of the bulb:

(1) increases

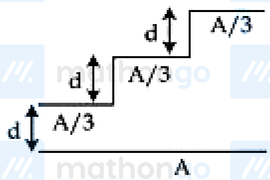
(2) decreases

(3) remains same

(4) becomes zero

Q7 - 2024 (09 Apr Shift 1)

A capacitor is made of a flat plate of area A and a second plate having a stair-like structure as shown in figure. If the area of each stair is $\frac{A}{3}$ and the height is d, the capacitance of the arrangement is :



(1) $\frac{13\epsilon_o A}{17d}$

(2) $\frac{11\epsilon_o \mathrm{A}}{18\ \mathrm{d}}$

(3) $\frac{18\epsilon_o \mathrm{A}}{11\ \mathrm{d}}$

(4) $\frac{11\epsilon_o \mathrm{A}}{20\ \mathrm{d}}$

Answer Key

Q1 (750) **Q2** (86) **Q3** (4) **Q4** (93)

Q5 (4) **Q6** (1) **Q7** (2)

Q1

Before inserting dielectric capacitance is given $C_0 = 12.5\text{pF}$ and charge on the capacitor $Q = C_0 V$ After inserting dielectric capacitance will become $\epsilon_r C_0$.

Change in potential energy of the capacitor

$$= E_i - E_f$$

$$= \frac{Q^2}{2C_i} - \frac{Q^2}{2C_f} = \frac{Q^2}{2C_0}\left[1 - \frac{1}{\epsilon_r}\right]$$

$$= \frac{(C_0 V)^2}{2C_0}\left[1 - \frac{1}{\epsilon_r}\right] = \frac{1}{2}C_0 V^2\left[1 - \frac{1}{\epsilon_r}\right]$$

Using $C_0 = 12.5\text{pF}, V = 12\text{ V}, \epsilon_r = 6$

$$= \frac{1}{2}(12.5) \times 12^2\left[1 - \frac{1}{6}\right] = \frac{1}{2}(12.5) \times 12^2 \times \frac{5}{6}$$

$$= 750\text{pJ} = 750 \times 10^{-12}\text{ J}$$

Q2

In parallel combination : Potential difference is same across all

$$\text{Energy } = \frac{1}{2}(C_1 + C_2 + C_3) V^2$$

$$= \frac{1}{2}(25 + 30 + 45) \times (100)^2 \times 10^{-6} = 0.5 = E$$

In series combination: Charge is same on all.

$$\frac{1}{C_{equ}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{25} + \frac{1}{30} + \frac{1}{45}$$

$$\frac{1}{C_{equ}} = \frac{(18 + 15 + 10)}{450} = \frac{43}{450} \Rightarrow C_{equ} = \frac{450}{43}$$

$$\text{Energy } = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_2} + \frac{Q^2}{2C_3}$$

$$= \frac{Q^2}{2}\left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right]$$

$$\frac{(V \times C_{equ})^2}{2} \times \frac{1}{C_{equ}} = \frac{V^2 C_{equ}}{2}$$

$$\frac{(100)^2}{2} \times \frac{450}{43} \times 10^{-6}$$

$$\Rightarrow \frac{4.5}{86} = \frac{9}{x}E = \frac{9}{x} \times 0.5 \Rightarrow x = 86$$

Q3

$E=\frac{E_0}{3} \Rightarrow V=\frac{V_0}{3}$

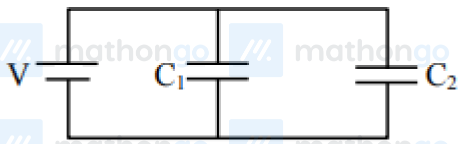
$\frac{V_0}{3}=V_0 e^{-\frac{t}{\tau}}$

$t=\tau \ell n 3$

$6.6 \times 10^{-6}=R\left(1.5 \times 10^{-6}\right)(1.1)$

$R=\frac{6}{1.5}=4 \Omega$

Q4



$C_{eq}=C_1+C_2$

$C_1=\frac{2 \epsilon_0 A}{2 \times d}=10 \mu F$

$C_2=\frac{3 \epsilon_0 A}{2 d}=15 \mu F$

$C_{eq}=25 \mu F$

Now the charge on

$C_1=10 V \mu c$

$C_2=1.5 V \mu C.$

Now force between the plates $\left[F=\frac{Q^2}{2 A \epsilon_0}\right]$

$\frac{100 V^2 \times 10^{-12}}{2 \times 2 \times 10^{-4} \epsilon_0}+\frac{225 V^2 \times 10^{-12}}{2 \times 2 \times 10^{-4} \times \epsilon_0}=8$

$325 V^2=8 \times 4 \times 10^{-4} \times 8.85$

$V^2=\frac{32 \times 8.85 \times 10^{-4}}{325}$

$\therefore V=\sqrt{\frac{283.2 \times 10^{-4}}{325}}$

$V=0.93 \times 10^{-2}$

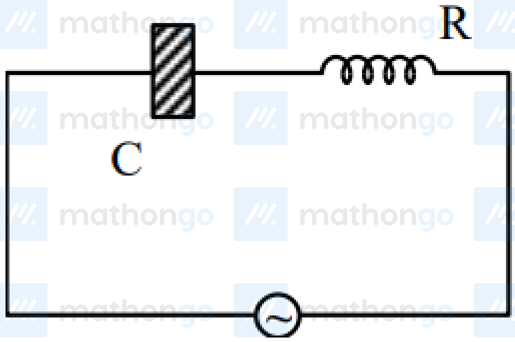
Q5

$$\frac{A\varepsilon_o}{d}=\frac{A\varepsilon_o}{\left(0.2+\frac{d}{k}\right)}$$

$$0.6=0.2+\frac{0.6}{k}$$

$$k=\frac{3}{2}$$

Q6



$Z=\sqrt{R^2+X_C^2} \& X_C=\frac{1}{WC}$

due to dielectric

$C \uparrow \Rightarrow X_C \downarrow \Rightarrow Z \downarrow$

So, current increases & thus bulb will glow more brighter.

Q7

All capacitor are in parallel combination.

Also effective area is common area only

$$\Rightarrow C_{eq}=C_1+C_2+C_3$$

$$\Rightarrow C_{eq}=\frac{A\varepsilon_0}{3d}+\frac{A\varepsilon_0}{3(2d)}+\frac{A\varepsilon_0}{3(3d)}$$

$$\Rightarrow C_{eq}=\frac{A\varepsilon_0}{3}\left(\frac{11}{6d}\right)$$

$$\Rightarrow C_{eq}=\frac{11A\varepsilon_0}{18d}$$